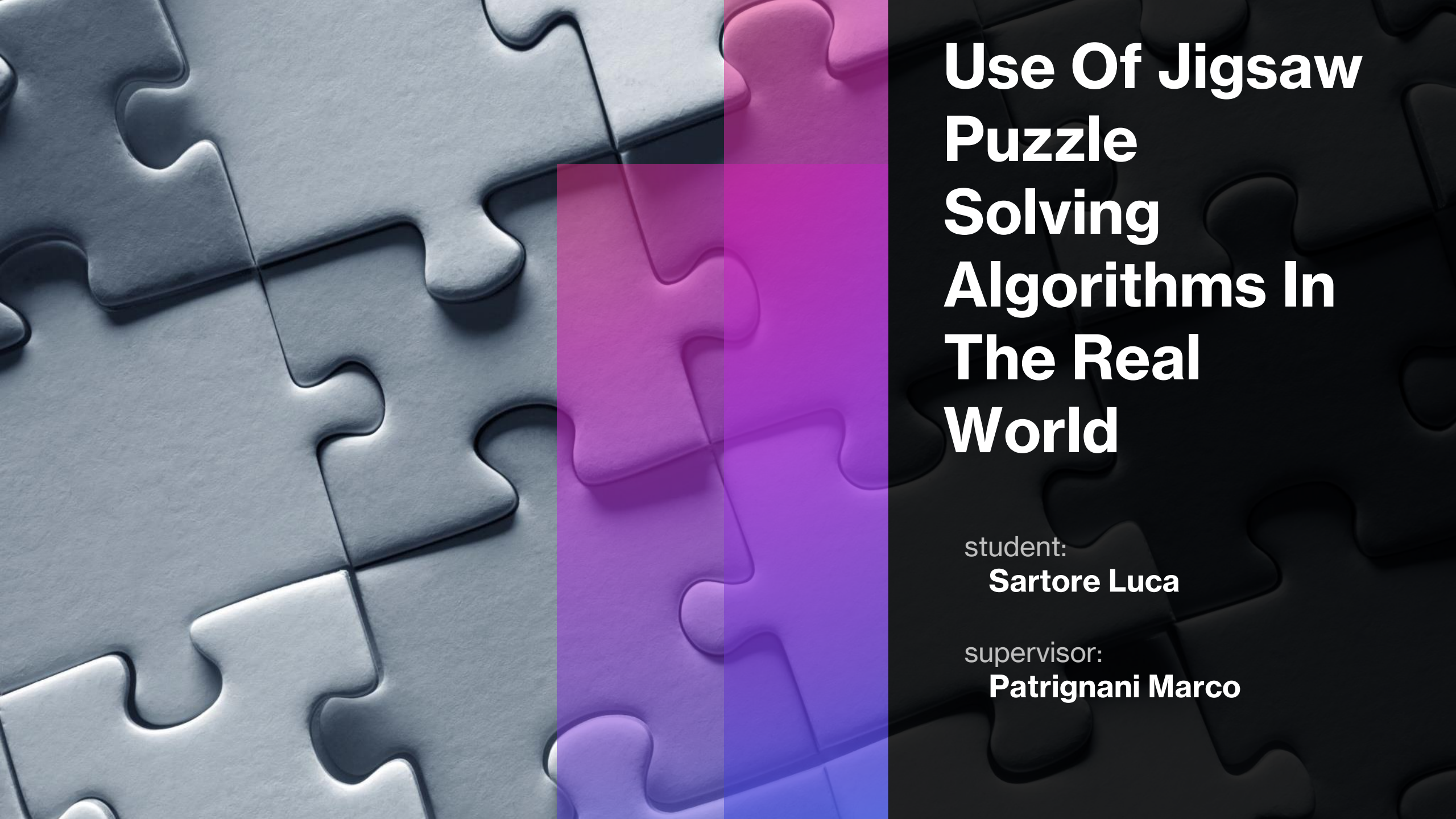




Use Of Jigsaw Puzzle Solving Algorithms In The Real World

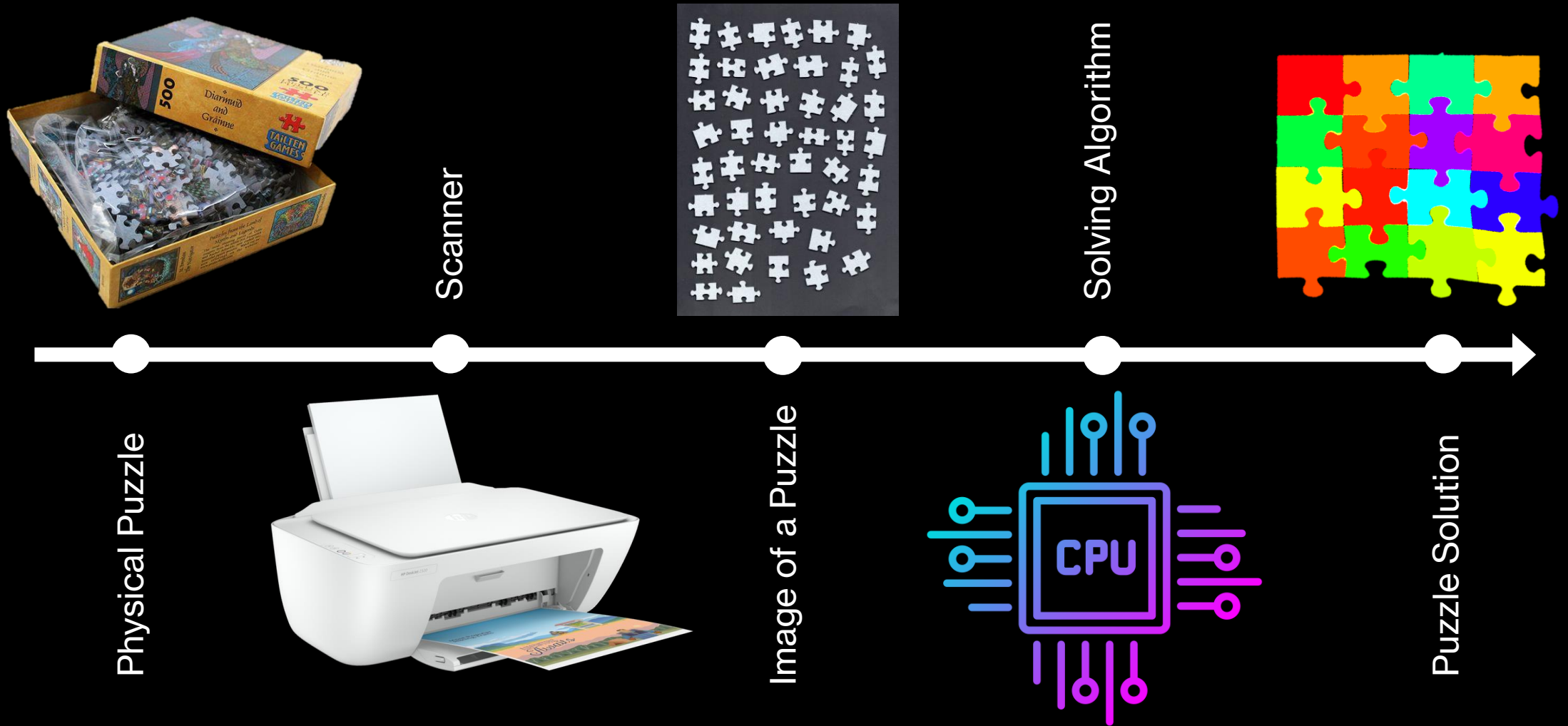
student:

Sartore Luca

supervisor:

Patrignani Marco

The Process



Two Kind of Jigsaw Puzzles

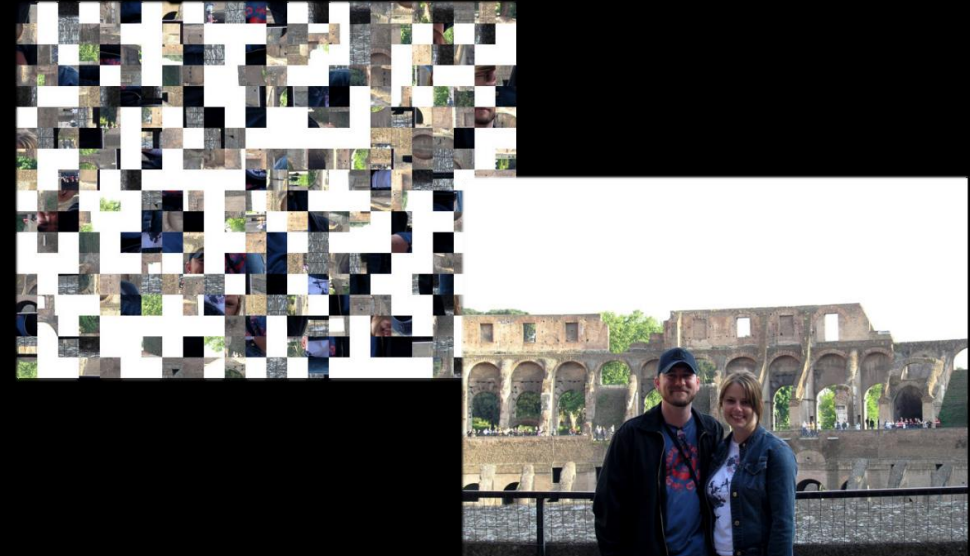
Real World Puzzles



Projects:

- AbtoSoftware, Computer Vision Powers Automatic Jigsaw Puzzle Solver, 2019

Digital Puzzles



Research Papers:

- Vahan Huroyan, Gilad Lerman and Hau-Tieng Wu, Solving Jigsaw Puzzles By The Graph Connection Laplacian, 2020.
- Dror Sholomon, Omid David and Nathan S. Netanyahu, A Genetic Algorithm-Based Solver for Very Large Jigsaw Puzzles, 2013

Same but Different

Inaccurate Representation



Pixel Perfect Representation

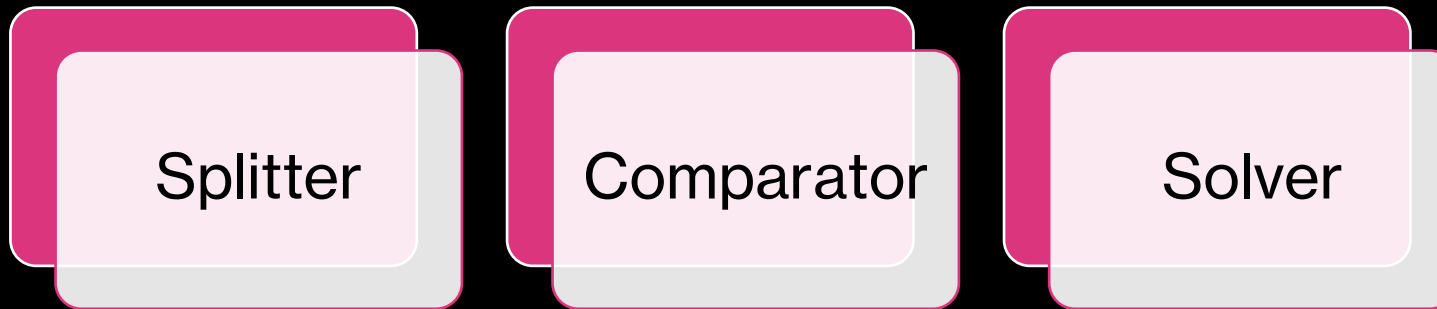


The Objectives



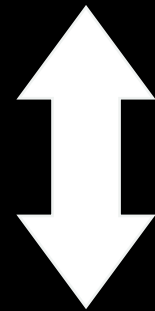
- **RQ1:** What is the difference in execution time between solving digital and real-world puzzles?
- **RQ2:** Is it possible to use the current state of the art solver for digital puzzles in the real world? What are the challenges to achieve this, and how can we overcome them?

Algorithm Structure



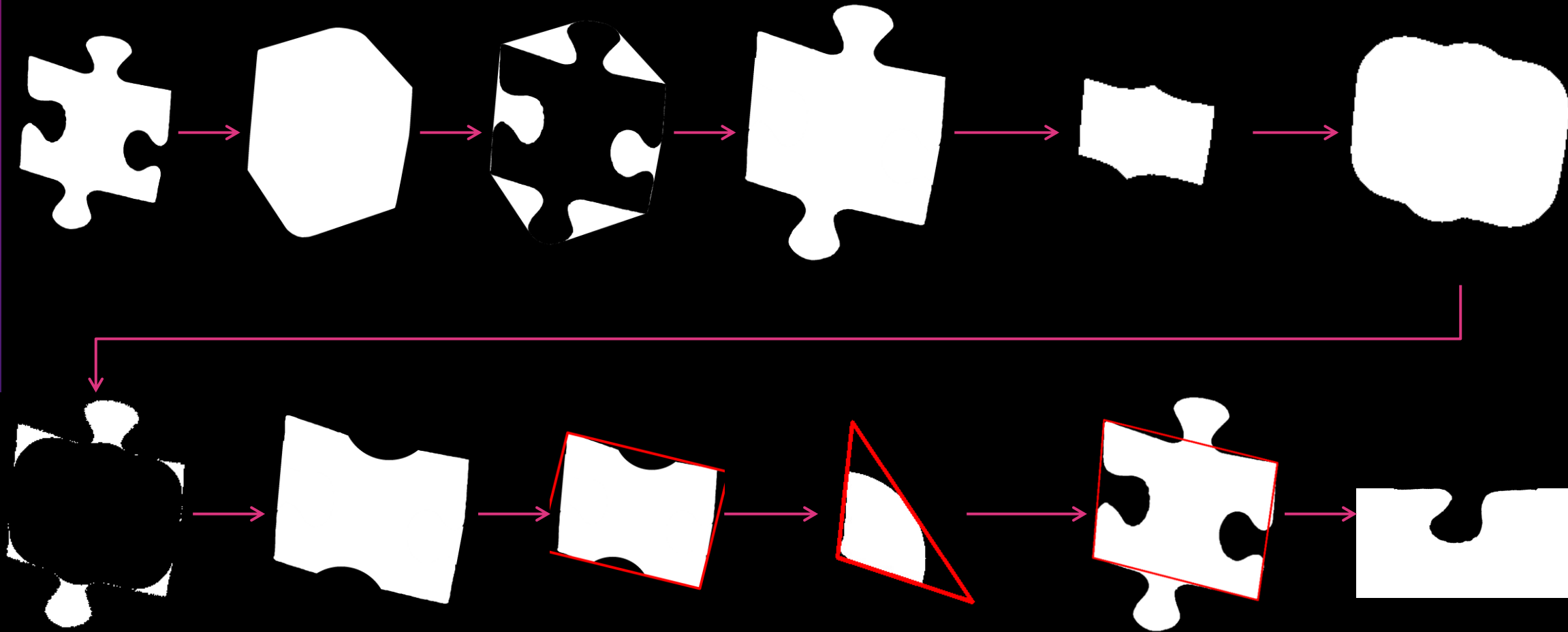
Riformulate RQ2

Is it possible to use the current state of the art solver for digital puzzles in the real world?



Can we build a comparator for real world puzzles that is accurate enough to work with digital puzzle solvers?

The Splitter

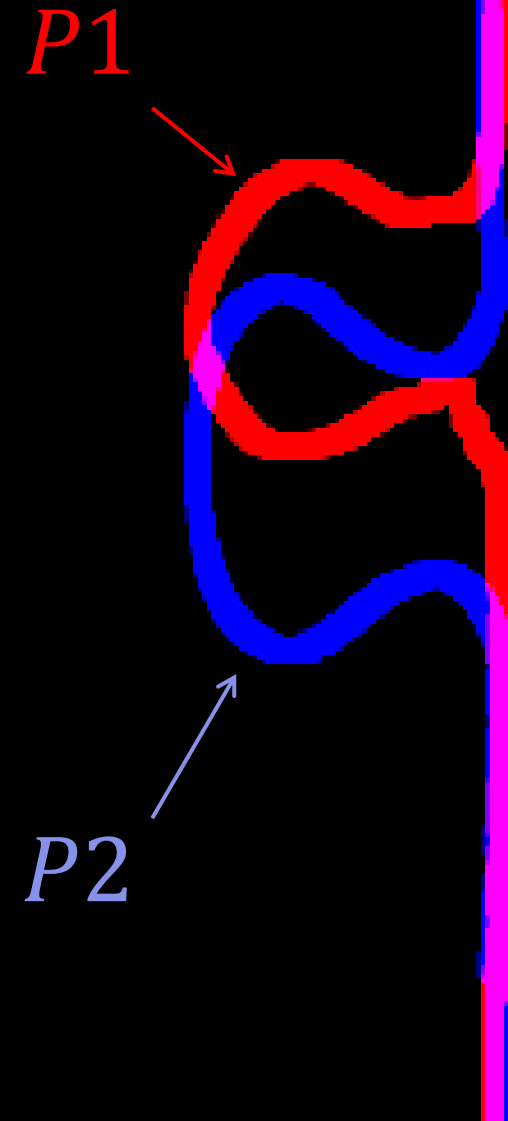


The Comparator

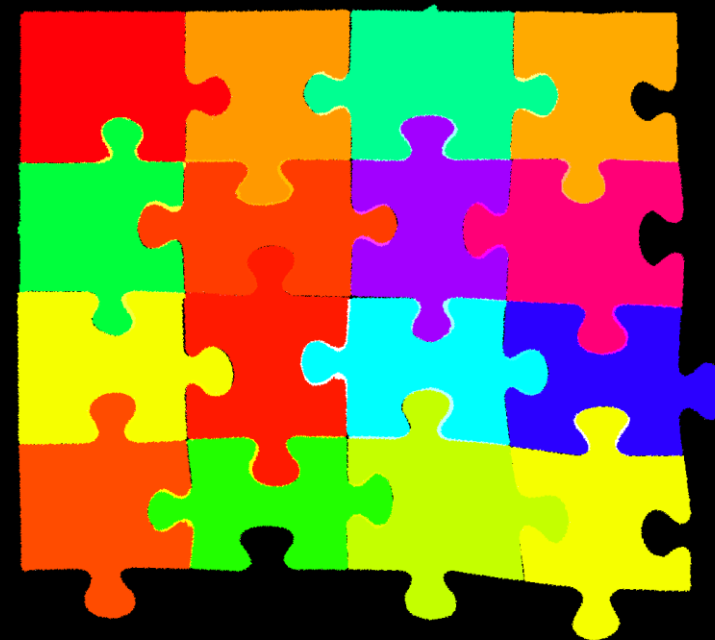
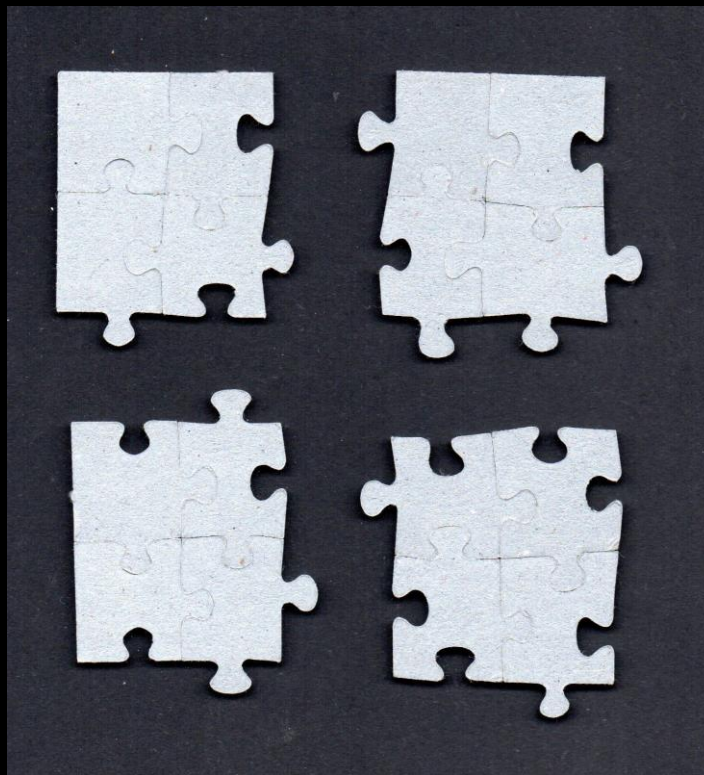
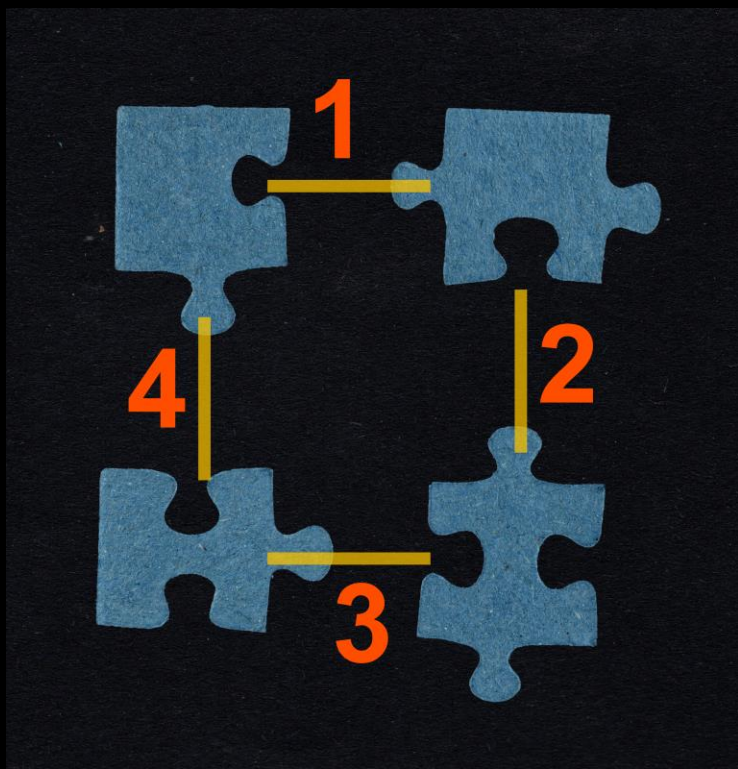
$$\text{Score} = \frac{\text{Area}(P1 \cap P2)}{\text{Area}(P1 \cup P2)}$$

$$\text{Score} \in \mathbb{R}$$

$$0 \leq \text{Score} \leq 1$$

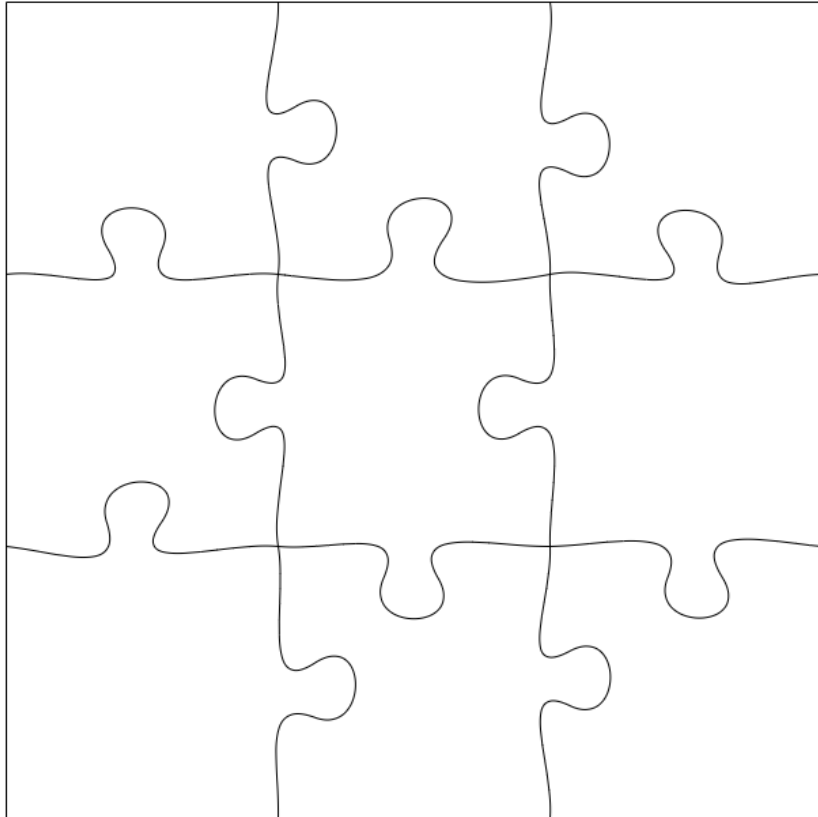


The Solver



Evaluation

Seed: 5668
Tab Size: 20%
Jitter: 4%
Tiles: 3 x 3
Size: 3 x 3 mm [Download SVG](#)



A digital puzzle to solve

<https://puzzle.telegnom.org/>

Execution Time vs Puzzle Size

RQ1: What is the difference in execution time between solving digital and real-world puzzles?

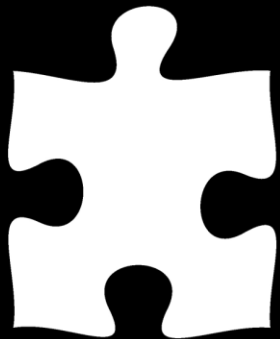
Size	Pieces	Execution time [s]	
		Digital puzzle	Real Puzzle
2x3	6	1	4
3x4	12	3	14
4x4	16	4	59
4x5	20	4	48
5x8	40	20	627
8x8	64	30	2851
8x12	96	660	N/A
16x16	256	377	N/A

Execution Time Consistency

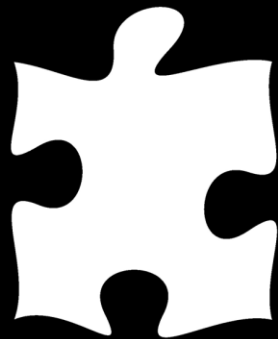
Run to run variance on a 4x4 real puzzle		
Execution time [s]	Average	Standard deviation
64	47.8	18.6
68		
43		
24		
29		
59		

Execution Time vs Jitter

Jitter = 4%



Jitter = 10%



Jitter effect on an 8x8 digital puzzle

Jitter	Execution time [s]
4%	104
5%	30
6%	17
8%	17
10%	16

Accuracy

Manually evaluated accuracy of the solution		
Puzzle size	Puzzle kind	Accuracy
4x4	real world	100%
8x8	real world	100%
16x16	digital	100%

Conclusions

Applying current State of the Art Algorithm to the Real World

RQ2: Can we build a comparator for real world puzzles that is accurate enough to work with digital puzzle solvers?

NO*

> See section 5.1 for full details

Improving the Comparator

Traditional computer vision

Pro:

- Can be developed relatively fast, without the need of a dataset

Cons:

- Hard to achieve high accuracy

Machine Learning

Pro:

- Good for achieving high accuracy even with imperfect data

Cons:

- Need a dataset that could require hundred of hours to collect

Labeling Tool

```
[  
  {  
    "piece_id": 12,  
    "side_0": {  
      "piece": 5,  
      "side": 0  
    },  
    "side_1": {  
      "piece": 8,  
      "side": 0  
    },  
    "side_2": null,  
    "side_3": null  
  }  
]
```





Thanks for your attention

Use Of Jigsaw Puzzle Solving Algorithms In The Real World

Sartore Luca



github.com/lucaSartore/PuzzleSolver

Questions?